Maximal Subgroups
Finitely Generated Groups and Zorn's Lemma

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Overview

Theorem

Nontrivial finitely generated subgroup possess maximal (proper) subgroups.

Definition

A subgroup is maximal if it is not contained strictly in a proper subgroup.

1. Proof
   - Definitions
   - Exercise §2.4: 17
   - Proofs
2. Discussion
   - Zorn's Lemma
   - Example
   - Questions

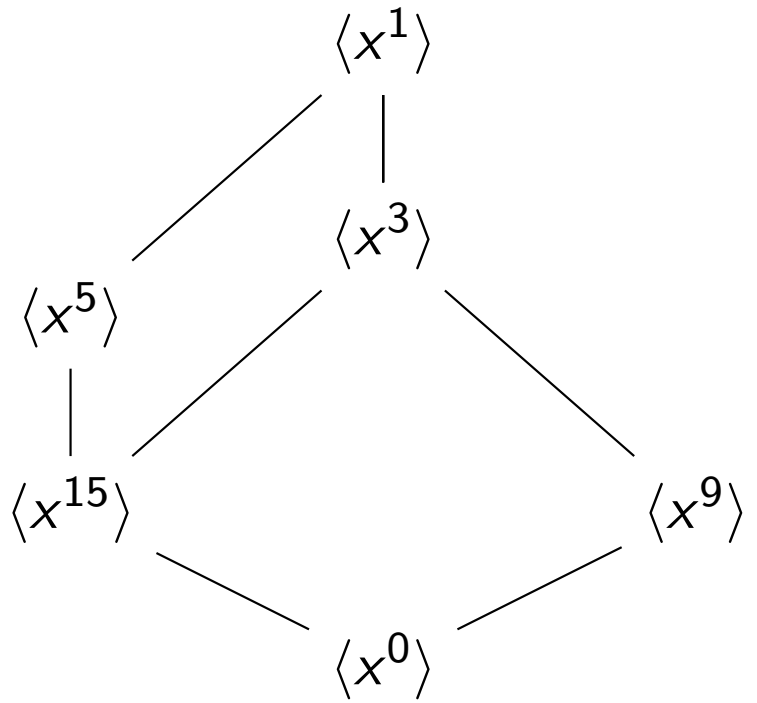


Figure: Maximal subgroups of $Z_{45} = \langle x^1 \rangle$ are $Z_{15} = \langle x^3 \rangle$ and $Z_9 = \langle x^5 \rangle$.

Definitions ([D&F, 2003] Appendix I)

Definition (poset)

A *partial order* on a nonempty set A is a relation $\leq$ on A satisfying

1. $x \leq x$ for all $x \in A$ (reflexive),
2. if $x \leq y$ and $y \leq z$ then $x = y$ for all $x, y \in A$ (antisymmetric),
3. if $x \leq y$ and $y \leq z$ then $x \leq z$ for all $x, y, z \in A$ (transitive).

Definition

Let the nonempty set A be partially ordered by $\leq$.

1. A subset $B \subset A$ is called a *chain* if for all $x, y \in B$, either $x \leq y$ or $y \leq x$.
2. An *upper bound* for a subset $B \subset A$ is an element $u \in A$ such that $b \leq u$, for all $b \in B$.
3. A *maximal element* of A is an element $m \in A$ such that if $m \leq x$ for any $x \in A$, then $m = x$.

Colloquialisms ([D&F, 2003] Appendix I)

- We usually say that A is a partially ordered set under the ordering $\leq$ or that A is partially ordered by $\leq$.
- In the literature a chain is also called a *tower* or called a *totally ordered* or *linearly ordered* or *simply ordered* subset.

Exercise D&S §2.4.17: Setup

- This is an exercise involving Zorn's Lemma (see Appendix I[1]) to prove that every nontrivial finitely generated group possesses maximal subgroups.
- Let G be a group, say $G = \langle g_1, g_2, \ldots, g_n \rangle$.
- Let $\mathcal{S} \subset \mathcal{P}(G)$ be the set of all proper subgroups of G. Then $\mathcal{S}$ is partially ordered by inclusion.
- Let $\mathcal{C}$ be a chain in $\mathcal{S}$.

[1] We will assume Zorn's Lemma in our proof, and discuss it afterward.

Exercise [D&F, 2003] §2.4: 17: Statements

Exercise 17 (a)

Prove that the union, H, of all the subgroups in $\mathcal{C}$ is a subgroup of G.

Exercise 17 (b)

Prove that H is a *proper* subgroup. [If not, each g_i must lie in H and so must lie in some element of the chain $\mathcal{C}$. Use the definition of a chain to arrive at a contradiction.]

Exercise 17 (c)

Use Zorn's Lemma to show that $\mathcal{S}$ has a maximal element (which is, by definition, a maximal subgroup.)

Exercise [D&F, 2003] §2.4: 17: Proof of (a)

Proof Setup (Hernandez) ...

Let $\mathcal{C}$ be a chain in the set $\mathcal{S}$ of all proper subgroups partially ordered by inclusion of a group $G = \langle g_i \rangle_{i=1}^{n}$ generated by a finite number $n \in \mathbb{N}$ of generators.
Consider two elements of the union of all the subgroups in $\mathcal{C}$:

$$x, y \in H = \bigcup_{H_\alpha \in \mathcal{C}} H_\alpha. \tag{1}$$

Then, there exist subgroups $H_1, H_2 \in \mathcal{C}$ such that $x \in H_1$ and $y \in H_2$. By the definition of a chain, we assume without loss of generality that $H_1 \leq H_2$.

(a) Then, H is a group because it contains the following:

$$xy, e, x^{-1} \in \langle x, y \rangle \subset H_2 \subset H \subset G. \tag{2}$$

Exercise [D&F, 2003] §2.4: 17: Proof of (b)

Proof (Hernandez).

(b) Suppose that the subgroup H is not *proper*, namely that $H = G$. Since $g_i \in H$, there exist subgroups $H_i \in \mathcal{C}$ such that $g_i \in H_i$. By the definition of chain, there exists a permutation $\sigma \in S_n$ such that $H_{\sigma(i)} \leq H_{\sigma(j)}$ for $1 \leq i \leq j \leq n$, i.e.:

$$H_{\sigma(1)} \leq H_{\sigma(2)} \leq \cdots \leq H_{\sigma(m)} \tag{3}$$

Then, each $g_i \in H_{\sigma(n)}$, so any $g \in G$ is in $H_{\sigma(n)}$, which contradicts its propriety. Therefore, H is a proper subgroup.

Exercise [D&F, 2003] §2.4: 17: Proof of (c)

Proof (Hernandez).

(c) According to the previous parts, $H_\alpha \leq H$ for all $H_\alpha \in \mathcal{C}$. By definition, H is an upper bound for $\mathcal{C} \subset \mathcal{S}$. By Zorn's Lemma (stated below), $\mathcal{S}$ has a maximal element $M \in \mathcal{S}$. Therefore, G has a proper maximal subgroup $M \leq G$.

Lemma (Zorn's Lemma)

If A is a nonempty partially ordered set in which every chain has an upper bound then A has a maximal element.

Joke (Bona [Krantz, 2002])

The Axiom of Choice is obviously true, the well-ordering principle obviously false, and who can tell about Zorn's lemma?

Discussion of Zorn's Lemma ([D&F, 2003] Appendix I)

Axiom (Axiom of Choice)

The Cartesian product of any nonempty collection of nonempty sets is nonempty. In other words, if I is any nonempty (indexing) set and A_i is a nonempty set for all $i \in I$, then there exists a choice function from I to $\cup_{i \in I} A_i$, i.e. a function:

$$f : I \to \bigcup_{i \in I} A_i,$$

such that $f(i) \in A_i$ for all $i \in I$.

Principle (The Well Ordering Principle)

Every nonempty set A has a well ordering, i.e. a total ordering on A such that every nonempty subset of A has a minimum (or smallest) element, i.e., for each nonempty $B \subset A$ there is some $s \in B$ such that $s \leq b$, for all $b \in B$.

Discussion of Zorn's Lemma ([D&F, 2003] Appendix I)

Theorem

Assuming the usual (Zermelo-Fraenkel) axioms of set theory, the following are equivalent:

1. Zorn's Lemma
2. the Axiom of Choice
3. the Well Ordering Principle

Theorem

Zorn's Lemma is independent of the usual (Zermelo-Fraenkel) axioms of set theory in the sense that including it (xor its negation) would preserve the consistency of the axioms of set theory (if ZF is indeed consistent).

Example

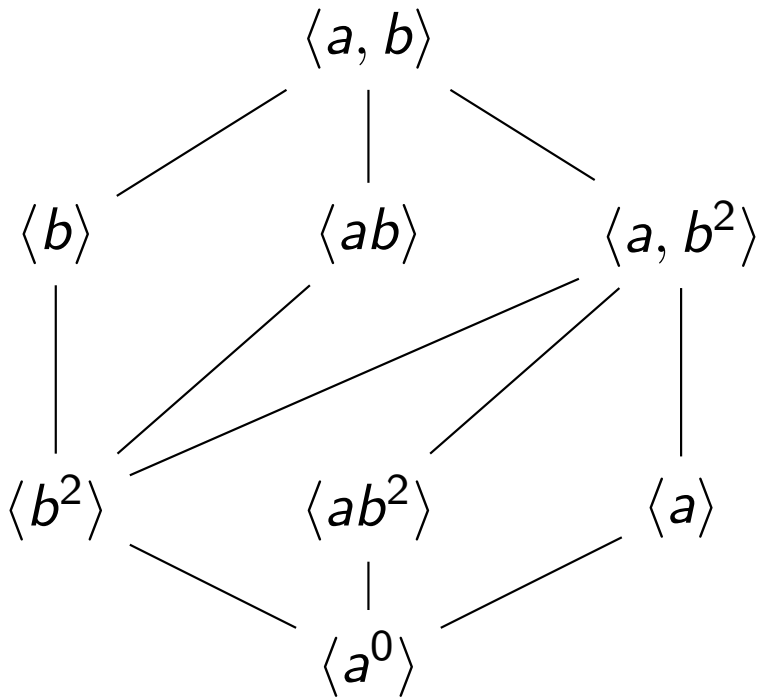


Figure: Subgroups of $Z_2 \times Z_4 = \langle a, b \rangle$

Remaining Questions

Question

When does a finitely generated group G have a *unique* maximal subgroup?

Proposition

This occurs when G is also simple, i.e. $G \cong Z_p$ for some prime p.

Question

When does G have not have any maximal subgroups?

Proposition

This occurs when G is divisible.

References



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Exercise §2.4.17 Solution.

https://mathemonads.github.io/assets/docs/631_2.4.17_oih.pdf.



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The Axiom of Choice, pp. 121–126

Handbook of Logic and Proof Techniques for Computer Science, Springer (2002).